

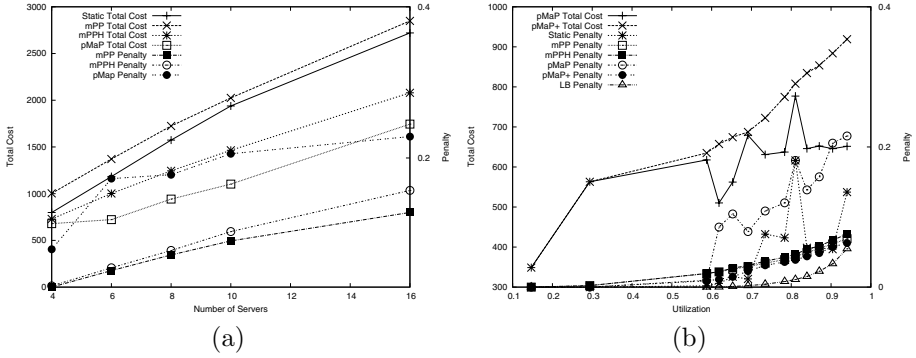


Fig. 9. (a) Overall Cost and Penalty with increase in servers (b) Power and Penalty with increase in fragmentation for penalty aware and unaware algorithms

mPPH are the top two performing algorithms even with increased number of servers. However, we observed (Fig. 9(a)) that while aiming to unbalance load for minimizing power, *pMaP* was leading to a large number of requests being dropped (or delayed). We call the drop in requests as *Penalty* of a placement. A reason for this penalty is the huge variance in workload. Hence, at an average load of 0.7 approximately, there are many instances when the transient load goes beyond 1.0 and, as a result, many requests need to be dropped (irrespective of the placement methodology). However, since *pMaP* is unaware of penalties, it drops more requests than others, while striving to minimize cost. This makes *pMaP* undesirable for use at high load intensities.

We next engineered *pMaP* to take penalty into account, while exploring the search space between the old placement and the new placement. The modified algorithm *pMaP+*, during its exploration, picks only those intermediates whose penalty are below a bound. We fixed the bound to be 0.1 in our experiments. We observe (Fig. 9(b)) that the penalty aware algorithm *pMaP+* now incurs the least penalty amongst *mPP*, *mPPH*, *pMaP* and static. The penalty seen by the Load Balanced algorithm is the baseline penalty incurred solely because of traffic variations and *pMaP+* now has a penalty approaching that of Load Balanced. This engineering allows *pMaP+* to again be the algorithm of choice under all scenarios. We also observe that the penalties of the *Static* placement also varies a lot, approaching 20% in certain cases. This is again, a result of the load-shifts that happen with time, re-affirming the importance of a dynamic placement strategy.

In this set of experiments, we simulate heterogeneity with various power models. We simulate another server with a power model where the static power or the dynamic power or both may be reduced to half. We increase the number of such servers from 0 to 2 to increase the complexity in Static Power or Dynamic Power (Fig. 10(a)). We observed that increased complexity (due to heterogeneity) leads to higher power savings relative to the Static Placement, which is the best power-unaware placement method. Further, the power savings are almost